DO NOW — RECALL

Lesson 113 · Open Source and Closed Source Decisions

Name: _____ Class: _____ Date: _____

1. Name a utility for recovering files after hardware loss.

2. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

3. What is the time complexity of sequential search in the worst case?

4. After pushing A then B onto a stack, which is popped first?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Backup and restore utility.

2. Model answer

X is removed; Y remains at the front.

3. Model answer

O(n).

4. Model answer

B, because a stack is LIFO.

By the end, you can…

• Open-source licences allow access to source code with specified reuse rights.

• Closed-source vendors control release; support and integration vary by product.

• Evaluate total cost, auditability, maintenance, expertise and obligations.

Vocabulary — know these terms

open source — Code distributed under a licence allowing inspection and specified reuse.

closed source — Source and modification rights normally controlled by owner.

licence — Legal conditions attached to software use.

vendor lock-in — Difficult or expensive dependence on one supplier.

NEW LEARNING

Read, annotate and explain

Source availability

Open-source licences allow access to source code with specified reuse rights.

Control vs support

Closed-source vendors control release; support and integration vary by product.

Make a justified choice

Evaluate total cost, auditability, maintenance, expertise and obligations.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A council wants to adapt an accessible web service using its development team. Explain an open-source advantage and a risk.

Model answer

Relevant open-source licence can permit modification and inspection, allowing accessibility changes. Risk: the council must ensure support, security patching and licence compliance; source availability alone guarantees neither quality nor safety.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define open-source and closed-source software and state a licensing implication. [6]

Question 2

Explain why free-of-charge software is not necessarily open source. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the choice for a charity's public website considering skills, privacy, costs and vendor dependence. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Open-source software is automatically more secure than closed-source software.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.